

Topic P7: Energy

P7.1 Work done

Summary

We now move on to consider how energy can be stored and transferred. This topic acts to consolidate the ideas of energy that have been covered in previous topics as it is a fundamental concept that underpins many of the ways in which matter interacts.

Underlying knowledge and understanding

Learners may have prior knowledge of energy listed as nine types, as this is the teaching approach often taken at Key Stage 2 and Key Stage 3 to increase accessibility to an abstract concept. Learners may find it difficult to move away from this idea but need to be able to approach systems in terms of energy transfers and stores. They will have an understanding that energy can be transferred in processes such as changing motion, burning fuels and in electrical

circuits. Learners should also be aware of the idea of conservation of energy and that it has a quantity that can be calculated.

Common misconceptions

Learners may have misconceptions around energy being a fuel-like substance that matter has to 'use up', that resting objects do not have any energy and that all energy is transferred efficiently. There is also often confusion between forces and energy.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
P7.1a describe for situations where there are energy transfers in a system, that there is no net change to the total energy of a closed system (qualitative only)	the law of conservation of energy			
P7.1b describe all the changes involved in the way energy is stored when a system changes for common situations	an object projected upwards or up a slope, a moving object hitting an obstacle, an object being accelerated by a constant force, a vehicle slowing down, bringing water to a boil in an electric kettle		WS1.2a, WS1.2b, WS1.3c, WS1.3f, WS1.4a, WS1.4e, WS2a, WS2b, WS2c	Exploring energy stores and transfers in different objects in a circus based activity. Objects could include a wind up toy, a weight on a spring, a weight being lifted or dropped, water being heated, electrical appliances

P7.1c	describe the changes in energy involved when a system is changed by heating (in terms of temperature change and specific heat capacity), by work done by forces, and by work done when a current flows				
P7.1d	make calculations of the energy changes associated with changes in a system, recalling or selecting the relevant equations for mechanical, electrical, and thermal processes; thereby express in quantitative form and on a common scale the overall redistribution of energy in the system	work done by forces, current flow, through heating and the use of kW h to measure energy use in electrical appliances in the home	M1a, M1c, M3c	WS1.3a, WS1.3b, WS1.3c, WS1.3e, WS2a, WS2b	Use of a joulemeter to measure the energy used by different electrical appliances. (PAG P5)
P7.1e	calculate the amounts of energy associated with a moving body, a stretched spring and an object raised above ground level		M1a, M1b, M1c, M2a, M3a, M3b, M3c, M3d	WS1.1b, WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3b, WS1.3c, WS1.3e, WS2a, WS2b	Use of light gates and trolleys to investigate kinetic energy. (PAG P5) Use of a joulemeter and electrical motor to lift a weight to investigate potential energy. (PAG P5) Investigation of energy changes and efficiency of bouncy balls. (PAG P5)